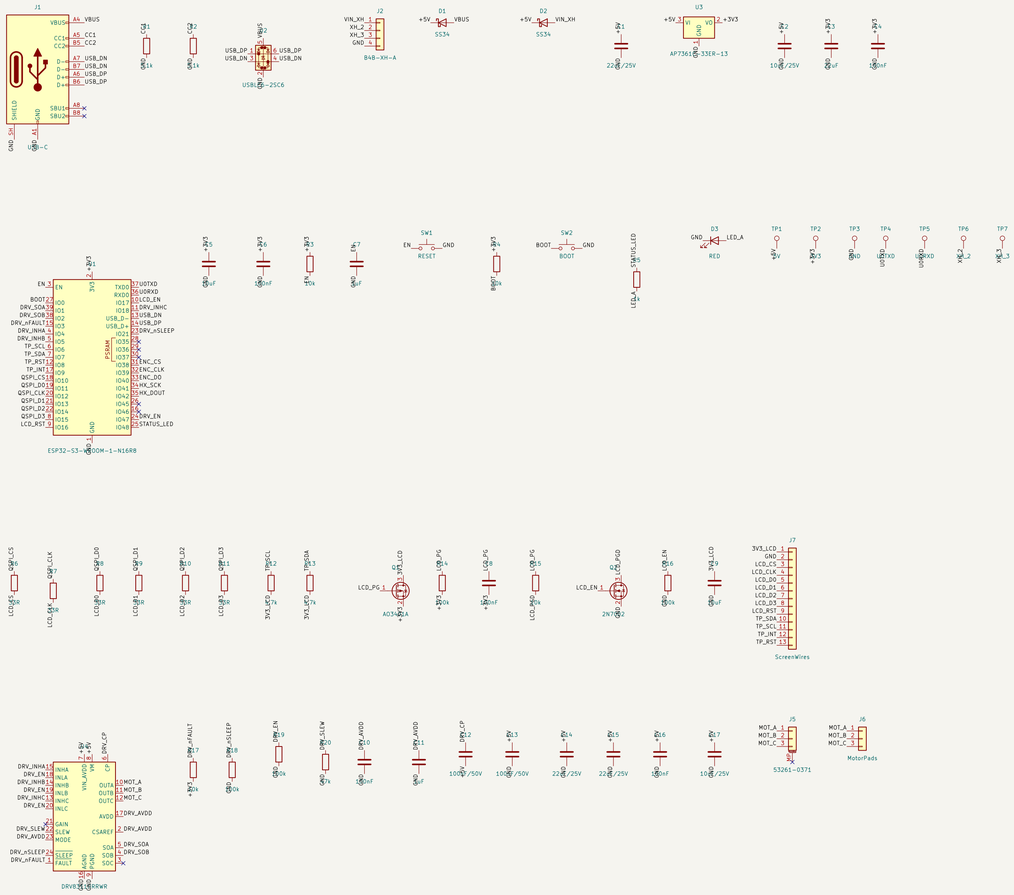


Design notes (board B):
- Power: USB-C VBUS and the GaggiMate cable (J2 pin1 +5V, pin4 GND, same pinout as the controller J4 display connector) are OR-ed by two SS34 Schottkys into +5V (~4.6 V). +5V feeds DRV8311H VM/VIN_AVDD and the AP7361C-33ER LDO.
- DRV8311H hardware variant: MODE tied to AVDD = 3x PWM mode (OCP 6 A). INHx = PWM from ESP32, all INLx tied to DRV_EN (GPIO47, 100k pull-down -> outputs Hi-Z at boot). nSLEEP GPIO21 with 100k pull-down. GAIN pin open = Hi-Z = 1 V/A CSA gain (+/-1.65 A range). SLEW 47k to GND = 75 V/us (dead time per datasheet). SOA/SOB -> GPIO1/GPIO2 (ADC1).
- Panel QSPI lines have 33R series resistors at the ESP32 (wire bundle ~10-15 cm long).
- Panel 3V3 (3V3_LCD) via AO3401A: firmware should keep LCD/TP pins low or Hi-Z while LCD_EN is low.
- MT6701 SSI lines (ENC_CS/CLK/DO) have 2.2k pull-ups here (board A has none), as SmartKnob.
- J7 (13 pads) and J6 (motor pads) are bare pads, not populated. Test pads TP1-TP7 are bare pads.

Panel 3V3 switched by P-FET (LCD_EN high = on, ~1 ms soft start). Touch I2C pull-ups to 3V3_LCD.
GPIO map = docs/superpowers/specs/2026-11-07-smartknob-input-design.md section 5.
ESP32-S3-WROOM-1-N16R8 + DRV8311H (hardware mode, 3x PWM) + USB-C/JST-XH 5 V diode-OR

GaggiMate Knob

Sheets: /
File: knob-main.kicad_sch

Title: GaggiMate Knob — board B (main: ESP32-S3, DRV8311H, power)

Size: A1	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 1/1